Hao (Edison) Wu

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Education

University of Wisconsin–Madison — Madison, WI
B.S., double major in Data Science and Information Science — *Sep. 2024 – Dec. 2026 (expected)*

- **Cumulative GPA: 4.00/4.00.**
- Selected coursework: Machine Learning; Text Mining; Big Data Systems; Database Design; Data Science Modeling; Statistical Data Visualization; Human Computer Interaction.

Xi'an Jiaotong–Liverpool University — Suzhou, China
Undergraduate study in Financial Mathematics; transferred to UW–Madison — *2022 – 2024*

Research Focus

My research focuses on reliable and efficient foundation models, particularly uncertainty quantification, adaptive inference, and multimodal reasoning. I investigate how models evaluate evidence, allocate inference-time computation, and improve prediction and generation across language, vision, and scientific tasks.

Submitted and Under-Review Manuscripts

[1] **Hao Wu**, Sipeng Chen, Qiancheng Yang, Hang Zhou, Xiao Luo. "Towards Realistic Evaluation of Uncertainty Quantification for LLM-Based Recommender Systems." *Transactions on Machine Learning Research (TMLR)*, 2026. **Under review; major revision completed.** *First author.* [OpenReview]

[2] **Hao Wu**, Sipeng Chen, Qiancheng Yang, Jingwen Fu, Yuwang Wang. "Closing Performance Gap for Diffusion Language Models under Low Inference Budgets." *ICLR 2027*. **Under review.** *First author.* [OpenReview]

[3] **Hao Wu**, Sipeng Chen, Qiancheng Yang, Hang Zhou, Xiao Luo. "CENTRE: Calibrated Response Trigger with Multi-resolution Retention for Online Video Understanding." *EACL 2027*. **Under review.** *Co-first author (first-listed).*

[4] Sipeng Chen[*], **Hao Wu**[*], Qiancheng Yang, Hang Zhou, Xiao Luo. "Opening-oriented Cost Regression with Multi-student Probing for LLM Thinking Budget Control." *Transactions of the Association for Computational Linguistics (TACL)*, 2026. **Submitted.** *Co-first author (second-listed); [*]equal contribution.*

Manuscripts in Preparation

[5] "Benchmarking Inference Strategies for Multi-Property Molecular Optimization." University of Wisconsin–Madison; research with Prof. Xiao Luo. **Full draft prepared; not submitted.** *Preparing for RECOMB 2027. Second author.*

[6] **Hao Wu**, Tianyi Chen, Sipeng Chen, Hang Zhou, Xiao Luo. "GOING: Generative Mode Switching with Confidence Gain for Real-World RAG Noise Mitigation." **Manuscript in preparation.** *Target venue: IEEE Transactions on Artificial Intelligence (TAI), 2026. First author.*

[7] Tianyi Chen, **Hao Wu**, TianMing Sha, Sipeng Chen, Junnan Li, Hang Zhou, Xiao Luo. "MARIE: Multimodal Routing with Evidence Token Grounding for 3D CT Report Generation." **Manuscript**, 2026. *Co-first author.*

Published Work

[8] Mingxuan Zhao, **Hao Wu**, Le Liu, Xiping Mao. "Forecast of Bond Issuance Based on ESG Score." *Academic Journal of Mathematical Sciences*, 4(3):52–59, 2023. *Corresponding author.* [DOI]

Honors and Awards

L&S Dean's List, University of Wisconsin–Madison — Fall 2024; Fall 2025; Spring 2026

University Excellence Award, Xi'an Jiaotong–Liverpool University — 2022 – 2024

Research Experience

University of Wisconsin–Madison Madison, WI
Undergraduate Research Assistant *Sep. 2025 – Present*
Research mentors: Prof. Xiao Luo (UW–Madison) and Prof. Hang Zhou (UNC–Chapel Hill)

- **RecUQ** *(first author):* Developed a fixed-ranking benchmark of eight UQ signals across four datasets: 42,475 eligible Llama records and a Qwen replication on 4,000 Warm-start cases. Distinguished failure detection from responses to controlled user-history and item-content changes.
- **CENTRE** *(first-listed co-first author):* Led problem formulation and methodology for frozen Video-LLMs, combining multi-resolution retention, horizon-aware response calibration, and order-robust answer readout.
- **MARIE** *(co-first author):* Contributed to evidence grounding for 3D CT report generation and implemented multimodal training and evaluation pipelines.
- **GOING** *(first author):* Proposed confidence-based generation control for retrieval-augmented generation, using relative confidence gain to identify when retrieved evidence degrades generation.
- **ORDER** *(co-first author, second-listed):* Developed shared-budget reasoning control using opening-based cost forecasts, a single continuation/restart priority queue, multi-student answer probing, and selective handover. Across six mathematics benchmarks, achieved 79.8% vs. 73.8% teacher-only macro accuracy under a teacher-token allowance of 4,694 per problem, and 86.3% vs. 82.0% under an allowance of 14,012.
- **Multi-property molecular inference benchmark** *(second author):* Compared nine inference strategies on two fixed molecular LMs and ten tasks (3,905 task–input pairs). Evaluated twenty ordered outputs per input to separate candidate success coverage, selection order, molecular validity, and generation workload.

Tsinghua University Beijing, China / Remote
Research Assistant — Diffusion Language Models *May 2026 – Present*

- **CaRR — Commitment-aware Refinement and Recovery** *(first author):* Developed token commitment remasking and trajectory-based output recovery to improve editable diffusion decoding within an existing inference schedule.
- Improved average accuracy across six code and mathematics metrics by **7.44 percentage points** over native ME-DLM at a budget of **64 scheduled model forwards**; evaluated four matched inference budgets without additional backbone forwards.

Teaching and Mentoring

Data Science Peer Mentor UW–Madison, Department of Statistics
Undergraduate mentoring in machine learning, statistics, and programming *Jul. 2025 – Present*

- Support students through weekly office hours and project debugging sessions in Python and R.

Project and Professional Experience

MadData 2026 Hackathon — MadNote AI Platform Feb. 2026
Backend and AI Developer, dotData @ UW–Madison. Co-developed a paper discovery platform with FastAPI, BERT keyword extraction and a Mistral-7B + RAG assistant over 10,000+ academic abstracts.

Guo Dao Asset Management Co., Ltd. — Data Engineering Intern Jun. – Aug. 2025
Shanghai, China. Developed Python/SQL trading-data pipelines and supported feature engineering and backtesting.

Dong Hai Securities Co., Ltd. — Data Analyst Intern Jul. – Aug. 2023
Changzhou, China. Automated extraction and preprocessing of financial records for 500+ companies using Python and Pandas.

Technical Skills and Languages

Research and ML: PyTorch, Hugging Face Transformers; LLMs/VLMs, diffusion language models, uncertainty quantification, calibration, RAG, multimodal evidence grounding.

Programming and Data: Python, R, MATLAB, SQL; Pandas, NumPy, Scikit-learn, XGBoost, LightGBM; Spark, Kafka, HDFS, Cassandra, MySQL, MongoDB, BigQuery; Git, Linux.

Languages: Mandarin (native), English (fluent), Spanish (elementary).